Section 1: Meta Information

Open Project Title	What Explains the Rise of Indian Chess?
Group ID	25
Mentor Name	Mayank Jain
Group Members	Rane (25B0975), Dhruv Parsewar (25B0977), Karmanya Gupta (25B1069), Shiven (25B1089)
GitHub Repository	(1) Official Classroom50 repository: `https://github.com/cs215-cse-iitb/dai-aug-2026-openproject-group-25` (2) Our development repository, where the work was done: `https://github.com/gitbromamide/Team_25` Our work was developed in repository (2) and copied into repository (1), the official course repository, for submission.
Video Presentation	`https://drive.google.com/file/d/1LhMcq5jA7zQdkZic85oL27BwoGUO_-ef/view?usp=sharing`

Honor Code and Discussions

We did not consult anyone outside our group.

LLM Usage Links

An LLM (Claude) was used to draft the structure and wording of this report and the LaTeX template. https://claude.ai/share/6b7d4734-89f0-4ef3-9ac8-d0b2fe418f65

GitHub Handles and Initial Commit

Member	Roll No.	GitHub handle	Earliest commit
Rane	25B0975	https://github.com/gitbromamide	17 October
Dhruv Parsewar	25B0977	https://github.com/dhruvparsewar	18 October
Karmanya Gupta	25B1069	https://github.com/karmanya-g7	21 October
Shiven	25B1089	https://github.com/shiventoshniwal	17 October

Contribution Breakdown

Unequal (an LLM was used, so Equal does not apply). Roll No. 25B0975: 100%, Roll No. 25B0977: 100%, Roll No. 25B1069: 100%, Roll No. 25B1089: 100%, LLM: 50%.

Section 2: Technical Content

Core Problem Statement

India's rise in chess is widely reported as a "boom" (Chennai hosted the 2022 Olympiad; D. Gukesh became world champion in December 2024), but headlines do not say what drove it. Competing explanations include a gradual build-up over many years, a jump after online chess grew during the COVID-19 lockdowns, or simply more Indians being rated by FIDE. Our central question is: *is the growth in India's high-rated players a steady long-term trend or did it accelerate after specific events, and is India unusual compared with other federations?* We split it into four sub-questions.

1. **Pool or strength?** Is the growth in players rated 2300+ and 2500+ explained by a larger rated pool or by a higher share of strong players within it?
2. **Timing.** Does growth accelerate after March 2020, the 2022 Olympiad or the 2024 title, relative to a pre-existing trend?
3. **Comparison.** Is India's share of the world's strong players rising, and is this unusual compared with China, the USA and Russia?
4. **Formats.** Do standard, rapid and blitz lists tell the same story?

Data Source(s)

Provenance and access. Monthly FIDE rating lists (XML), `https://ratings.fide.com/download_lists.phtml`. They are public and need no login. FIDE's downloadable archive starts in Feb 2015 (our proposal said 2001; corrected at Checkpoint 1). **Scale.** We use the January and July lists for 2015–2026 plus the Oct 2026 list: 24 lists per format.

Format	Lists	Players per list	Total rows
Standard	24	214,727 – 569,654	9.12M
Rapid	24	51,100 – 479,701	5.47M
Blitz	24	41,785 – 317,994	3.83M

Structure. Each row has 7 source fields (FIDE ID, federation, sex, title, rating, birth year, flag) plus snapshot and format columns. We aggregate to one row per format, list date and federation, with 12 measures: rated players; players above 2000/2300/2500/2700; IM+GM and GM counts; mean, top-10 and top-100 ratings.
Cleaning: streaming XML parse to Parquet; format and date from file names; numeric coercion; unrated players excluded; inactive players flagged; non-XML lists skipped and logged.

Tools and Software Packages

Python: pandas, NumPy, SciPy (`scipy.stats`), matplotlib
Parquet storage; GitHub.

India Focus

The primary group is federation IND, i.e. records of Indian players, and the question concerns India's chess rise and events tied to India (the 2022 Olympiad, the 2024 world title). Other federations appear only as benchmarks to judge whether India's pattern is unusual.

Methods

(i) *Pool vs. strength:* compare growth of the rated pool with growth of 2300+/2500+ counts and with the 2300+ share of the pool.

(ii) *Timing:* fit a pre-2020 trend to log counts and check whether post-event points fall outside the prediction interval; as a robustness check, fit a segmented regression with a break term and report its confidence interval. With only two snapshots a year, we treat results as descriptive.
(iii) *Benchmarks:* India's share of the world's players above each threshold, compared with China, the USA and Russia.
(iv) *Formats:* repeat (i)–(iii) for rapid and blitz.

Challenges and Results Achieved

Challenges.

- *Archive start and sparse series:* FIDE's downloadable archive begins in Feb 2015, not 2001, and we use only January and July lists, so each series has 24 points. About 18.4M rows were parsed from XML.
- *Unstable active-player flag:* India's active count swings from 11,858 (Jan 2020) to 9,040, 12,451, 9,405 and 5,444 (Jan 2022) while the rated pool moves smoothly. We could not confirm the cause, so our main analysis uses all rated players.
- *List changes:* the rapid and blitz lists both step up between Jul 2022 and Jan 2023 (India's blitz 2300+ count 99 → 130, rapid 78 → 122), which looks like a change in how the lists work.
- *Threats to validity:* rating inflation, correlated snapshots and hand-picked comparison federations; we make no causal claims.

Results.

- *Pool vs. strength:* between Jan 2016 and Jan 2026 (standard) the rated pool grew 2.6× (20,123 → 51,859) but 2300+ players only 2.0× (141 → 281), so the 2300+ share fell from 0.70% to 0.54% (change −0.16 points, 95% CI −0.29 to −0.03). Over Jul 2015 to Oct 2026 the pool grew 3.1×, while 2300+ and 2500+ players each grew 2.5× and GMs 2.6×.
- *Timing:* the pool was nearly flat from Jan 2020 to Jan 2022 (32,730 → 33,800). The 2300+ count grew about 6.8% a year overall; growth slowed at COVID (−2.9 points, CI −4.6 to −1.1, $p = 0.003$), while the Olympiad (+0.9, $p = 0.40$) and Gukesh (+4.3, $p = 0.14$) changes are not distinguishable from 15 placebo break dates (range −4.6 to +4.3).
- *Benchmarks:* India's share of the world's 2300+ players rose from 1.5% to 3.7% and its GM share from 2.5% to 5.1%, while its share of all rated players rose only from 8.6% to 10.1%. India's 2300+ count grew about 8% a year (CI 6.2% to 9.8%) against about 3.5% for China and 1.8% for the USA.
- *Formats:* the 2300+ group grew 2.5× in standard (CI 2.0 to 3.0), 1.6× in blitz and 1.5× in rapid, with no jump after COVID in either; the pools grew far more in rapid (15×) and blitz (10×).

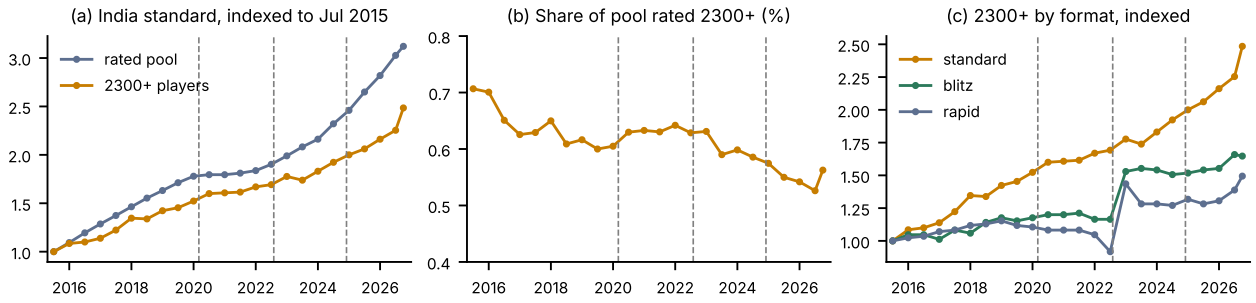


Figure 1: India, Jul 2015 to Oct 2026. Dashed lines: COVID (Mar 2020), Olympiad (Aug 2022), Gukesh's title (Dec 2024).

Section 3: Frequently Answered Questions (FAQ)

Q1 (objective). Is the growth in India's players rated 2300+ and 2500+ explained by a larger rated pool or by a higher share of strong players?

Mostly a larger pool. Between Jan 2016 and Jan 2026 (standard) the rated pool grew 2.6× (20,123 → 51,859) but 2300+ players only 2.0× (141 → 281), so the 2300+ share fell from 0.70% to 0.54%. A two-proportion interval puts the change at −0.16 percentage points (95% CI −0.29 to −0.03, $p \approx 0.02$); the ratio of 2300+ counts is 1.99 (95% CI 1.63 to 2.44, treating counts as Poisson). Had the share stayed at its 2016 level, 141 players would have become about 363, not 281. *Caveat:* the same players appear in both lists, so the samples are not independent and the intervals are a rough guide; FIDE's lower rating floor over these years may also add low-rated players to the pool.

Q2 (subjective). Does growth accelerate after March 2020, the 2022 Olympiad or Gukesh's 2024 title, relative to the pre-existing trend?

Not after COVID, where growth slowed; later speed-ups cannot be told apart from ordinary variation. The 2300+ count grew about 6.8% a year overall (95% CI 6.3% to 7.3%). Fitting a line to log(2300+) with a break at each event, the change in yearly growth is −2.9 points at COVID (CI −4.6 to −1.1, $p = 0.003$), +0.9 at the Olympiad (CI −1.2 to +3.0, $p = 0.40$) and +4.3 at Gukesh's title (CI −1.5 to +10.0, $p = 0.14$). As a placebo check, the same model at 15 arbitrary break dates gives changes from −4.6 to +4.3 points (sd 2.9), so the Gukesh value is at the top of that range, not beyond it. *Caveats:* we ran several tests, so one small p-value is not strong evidence; consecutive snapshots are correlated, which makes the intervals too narrow; and the events are close together and overlap with rule changes, so a break would not show which event caused it.

Q3 (objective). Is India's share of the world's strong players rising, and is this unusual compared with China, the USA, Russia and others?

Yes, among the six federations we compared. India's 2300+ count grew about 8.0% a year (95% CI 6.2% to 9.8%), against China 3.5% and the USA 1.8%, while Germany and Russia shrank slightly. India's share of the world's 2300+ players rose from about 1.5% to 3.7%. Strong players per rated player is still low in India (about 0.5%, versus about 4.8% for China and 3.7% for the USA), and that share fell in every federation we plotted, so the participation story is not only Indian. *Caveat:* the benchmark federations were chosen by us, and a different group could change the ranking.

Q4 (subjective). Do the standard, rapid and blitz lists tell the same story?

Same direction, different size. From Jul 2015 to Oct 2026 the 2300+ group grew 2.5× in standard (CI 2.0 to 3.0), 1.6× in blitz (CI 1.3 to 2.2) and 1.5× in rapid (CI 1.1 to 2.0), whereas the rated pools grew far more in rapid and blitz. Rapid and blitz strong-player counts were nearly flat from 2016 to 2022, so online chess shows no jump there either. *Caveat:* both lists step up between Jul 2022 and Jan 2023, which looks like a change in how the lists work as much as a change in players; and the rapid and blitz pools started small, which inflates their multiples.

Q5 (objective). How large is the dataset and is it freely accessible?

We use 24 lists per format (Jan and July 2015–2026 plus Oct 2026): 9.12M standard, 5.47M rapid and 3.83M blitz rows. The lists are public and need no login. FIDE's archive starts in Feb 2015, not 2001. The latest standard list (Oct 2026) has 569,654 rated players from 213 federations, and India has the most rated players of any federation (57,412, ahead of Russia with 44,115 and Spain with 40,225). Our audit found no duplicate FIDE IDs and birth year missing for 1.5% of players.

Q6 (methodological). Why use all rated players instead of only active ones?

Because the active count is not a stable measure of who plays. We read the `flag` field as marking inactive players (values `i`, `wi`) versus active ones, and call the result our "active guess". India's active count swings from 11,858 (Jan 2020) to 9,040 (Jul 2020), 12,451 (Jan 2021), 9,405 (Jul 2021) and 5,444 (Jan 2022), before recovering to 19,880 by Oct 2026, while the rated pool moves smoothly (32,730 → 33,800 over Jan 2020 to Jan 2022). We suspect the swings come from the pause in over-the-board play and FIDE's activity rules, but we could not confirm this. The rated pool avoids depending on the flag, at the cost of including some players who no longer play.

Q7 (subjective). What limits the conclusions?

(i) *Participation vs. strength:* a larger pool says little about playing strength. (ii) *Stale entries:* about 61% of players in the Oct 2026 standard list are flagged inactive, yet still counted as rated. (iii) *Rating inflation and rule changes:* rapid and blitz lists both step up between Jul 2022 and Jan 2023 (India's blitz 2300+ count 99 → 130, rapid 78 → 122), which looks like a change in how the lists work. (iv) *Sparse data:* two snapshots a year give 24 points per series, and consecutive points are correlated. (v) *Choice of comparison federations:* we picked China, the USA, Russia, Germany and France ourselves. The analysis is descriptive and does not establish causes.

Q8 (objective). Why use 2300, 2500 and 2700 as thresholds for "strong" players, and do they tell the same story?

They are standard milestones in FIDE ratings: 2300 is the level associated with the FIDE Master title, 2500 with the Grandmaster level (the GM title also needs norms), and 2700 marks the world's elite. For India (standard, Jul 2015 → Oct 2026) the 2300+ count grew 2.5× (130 → 323), the 2500+ count 2.5× (21 → 52) and GMs 2.6× (38 → 97), while players above 2000 grew only 1.4× (1,365 → 1,918) against a pool that grew 3.1×. The 2700+ group is too small to read (2 → 5, peaking at 7 in Jul 2025), and India's top-10 mean rating rose from 2656 to 2704, peaking at 2725 in Jul 2025.

Q9 (subjective). Can we say the "chess boom" was caused by online chess or by Gukesh's title?

No. Our data show when growth changed, not why. India's rated pool was nearly flat from Jan 2020 to Jan 2022 (32,730 → 33,800, +3%, after +22% in the previous two years) and then grew again (+18% from Jan 2022 to Jan 2024), so the only clear break in the exploratory data is a slowdown around COVID. India's share of the world's 2300+ players rose through most lists, from 1.5% (Jul 2015) to 2.3% (Jan 2020), 2.5% (Jan 2022) and 3.7% (Oct 2026), with no obvious jump at the Olympiad or the title. Online chess, the Olympiad, the title and rule changes overlap in time, so we describe timing, not causes.